

LIQUIFIED NATURAL GAS (LNG)

CHE 396 - SENIOR DESIGN I
PROJECT #1

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INTRODUCTION

The overall objective of our project is to focus on the underlying issues generated from the processing of Liquefied Natural Gas (LNG).



WHAT IS LIQUIFIED NATURAL GAS?

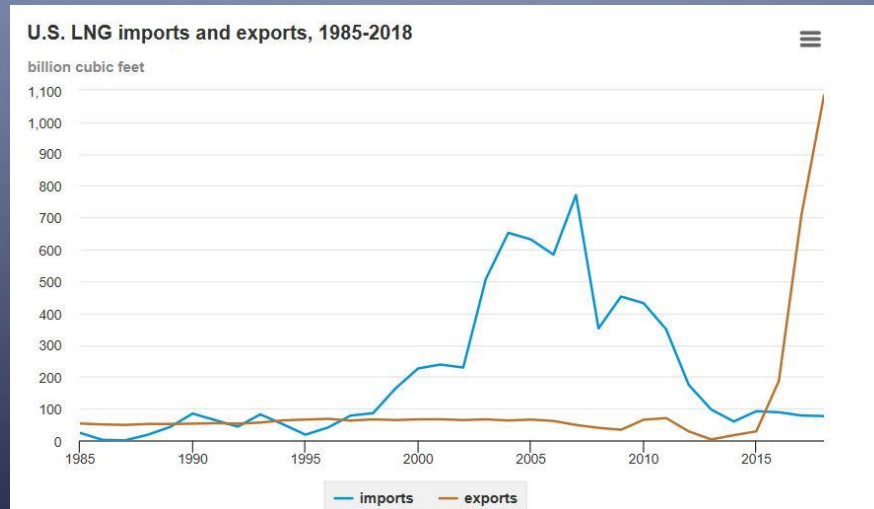
- Liquefied Natural Gas is a natural gas that has been cooled down to -260°F , in a liquid state.
- This allows gas to shrink 600 times less than it was in the gaseous state.
- Liquid natural gas is $> 90\%$ methane, the other 10% is light hydrocarbons (ethane, propane, butane)

WHY LNG?

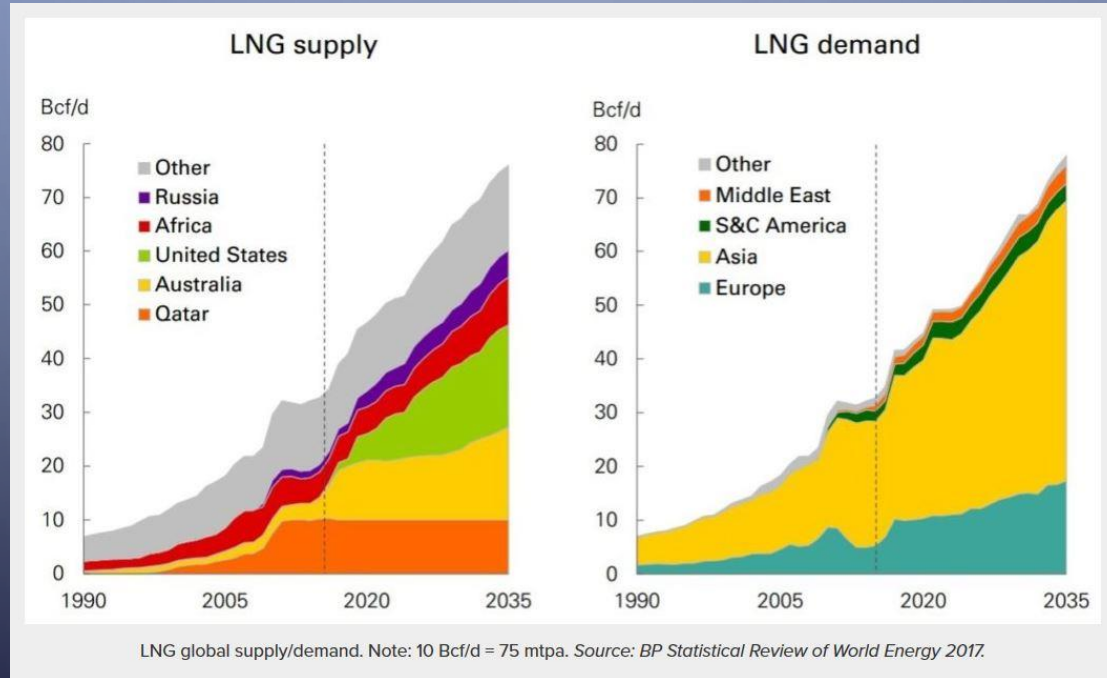
- LNG allows for more storage of natural gases in smaller tanks
- It is easier and safer to transport
- Higher Trade Relations between producers and consumers
- Cleaner than other fossil fuel
- High demand for industrial and residential use

Market Analysis

- U.S got the title of Net Exporter of LNG, averaging 4.2 Billion Cubic Feet a day of exports as March 2019.
- New export volume in Freeport terminal in Texas.



- New suppliers of LNG : Qatar (2010), Australia (2014), USA (2016).
- Decrease in prices



What does that mean for the U.S ?

- Expecting the creation of 3,000 engineering and construction jobs with Freeport -Tx alone
- Hundreds of indirect jobs
- \$73 billion by 2040.

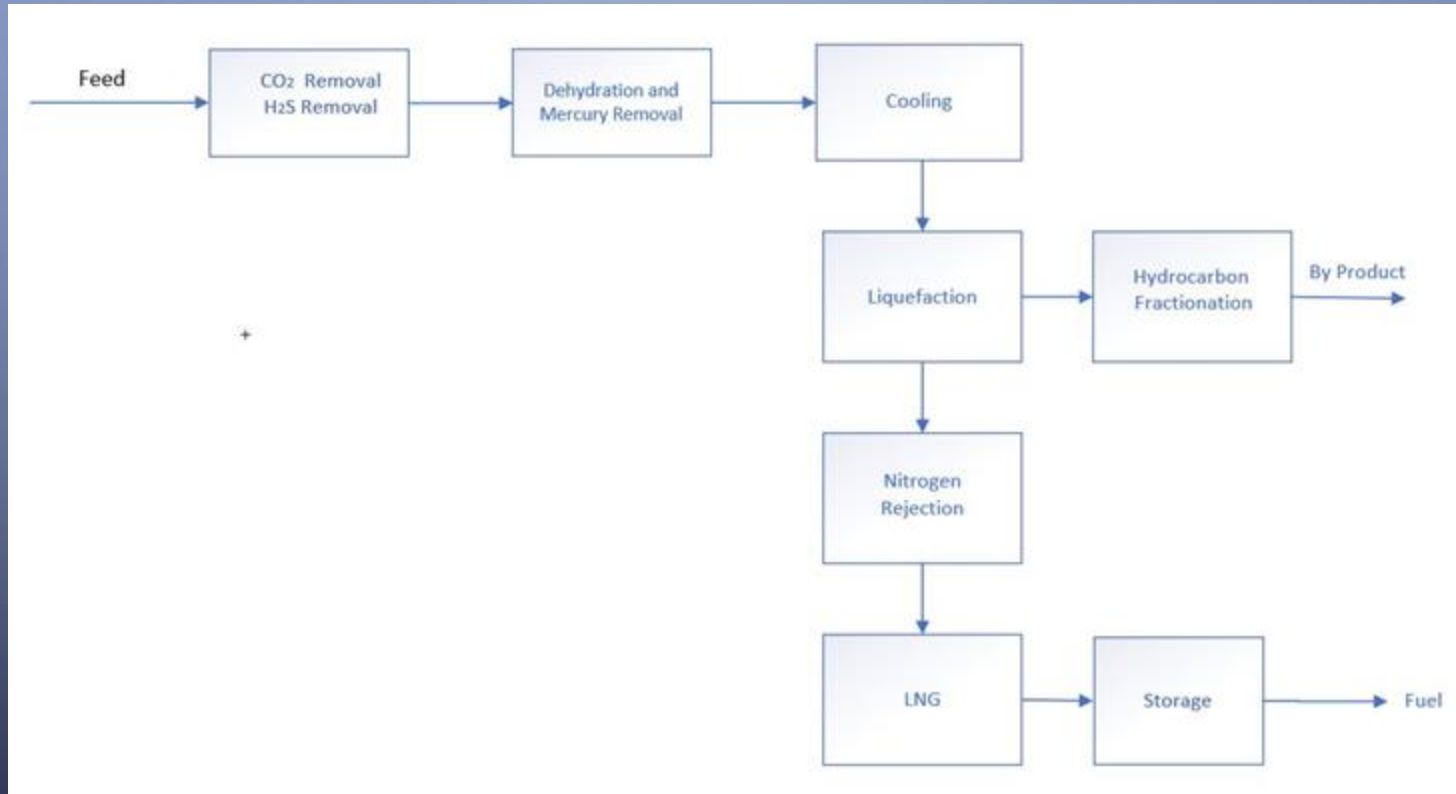
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Process Description

PROCESS DESCRIPTION

- Extraction of Impure Natural Gas from shale deposits
- Removal of impurities (N_2 , Hg, etc.) in sizes through pipes and vessels
- Absorption of CO_2 and H_2S by water-based solvent using separator
- Removal of remaining water from hydrocarbons through dehydration
- Precooling and removal of heavy hydrocarbons by fractination
- Liquefaction of pure methane using refrigeration and heat exchangers

BFD:



Acid Gas Removal			
Component	Units	IN	Out
Methane	mmcf/day	16.199	16.199
Ethane	mmcf/day	2.388	2.388
Propane	mmcf/day	0.646	0.646
Carbon Dioxide	mmcf/day	0.404	0.081
Butane	mmcf/day	0.224	0.224
Nitrogen	mmcf/day	0.038	0.038
Heavy Hydrocarbons	mmcf/day	0.100	0.100
Hydrogen Sulfide	mmcf/day	0.0008	0.000

Dehydration			
Component	Units	IN	Out
Methane	mmcf/day	16.199	16.199
Ethane	mmcf/day	2.388	2.388
Propane	mmcf/day	0.646	0.646
Carbon Dioxide	mmcf/day	0.404	0.081
Butane	mmcf/day	0.224	0.224
Nitrogen	mmcf/day	0.038	0.038
Heavy Hydrocarbons	mmcf/day	0.100	0.100
Hydrogen Sulfide	mmcf/day	0.0008	0.000

Cooling			
Component	Units	IN	Out
Methane	mmcf/day	16.199	16.199
Ethane	mmcf/day	2.388	2.388
Propane	mmcf/day	0.646	0.646
Carbon Dioxide	mmcf/day	0.404	0.081
Butane	mmcf/day	0.224	0.224
Nitrogen	mmcf/day	0.038	0.038
Heavy Hydrocarbons	mmcf/day	0.100	0.100
Hydrogen Sulfide	mmcf/day	0.0008	0.000

Liquefaction			
Component	Units	IN	Out
Methane	mmcf/day	16.199	16.199
Ethane	mmcf/day	2.388	0.725
Propane	mmcf/day	0.646	0.052
Carbon Dioxide	mmcf/day	0.404	0.081
Butane	mmcf/day	0.224	0.005
Nitrogen	mmcf/day	0.038	0.038
Heavy Hydrocarbons	mmcf/day	0.100	0.005
Hydrogen Sulfide	mmcf/day	0.0008	0.000

Hydrocarbon Fractionation			
Component	Units	IN	Out
Methane	mmcf/day	0.000	0.000
Ethane	mmcf/day	1.663	1.663
Propane	mmcf/day	0.594	0.594
Carbon Dioxide	mmcf/day	0.000	0.000
Butane	mmcf/day	0.219	0.219
Nitrogen	mmcf/day	0.000	0.000
Heavy Hydrocarbons	mmcf/day	0.095	0.095
Hydrogen Sulfide	mmcf/day	0.0000	0.000

Nitrogen Rejection			
Component	Units	IN	Out
Methane	mmcf/day	16.199	16.199
Ethane	mmcf/day	2.388	0.725
Propane	mmcf/day	0.646	0.052
Carbon Dioxide	mmcf/day	0.404	0.081
Butane	mmcf/day	0.224	0.005
Nitrogen	mmcf/day	0.038	0.008
Heavy Hydrocarbons	mmcf/day	0.100	0.005
Hydrogen Sulfide	mmcf/day	0.0008	0.000

Current Liquefaction

Most common LNG process cycles in industry:
Technologies/Cycles

1. *Single Mixed Refrigerant*

- consists of nitrogen and hydrocarbons in one mixed refrigerant loop
- Reversible operation possible due to small temperature gap, good thermodynamics efficiency, low power requirements
- Requires few machines, one compressor, small vessels for separation
- Only useful for medium scale operation due to moderate plant complexity

2. *Double Mixed Refrigerant*

- Utilizes two mixed refrigerant units that are used for pre-cooling and liquefaction
- Uses two compressor strings to ensure non stop train operation
- Requires multiple heat exchangers
- Interstage mixing of non-equilibrium streams in mixed refrigerant compression unit leads to thermodynamics irreversibilities (low cycle efficiency)

3. *Propane Mixed Refrigerant*

- Most popular method for onshore liquefaction plants
- Similar to SMR but uses propane as precoolers with lower molecular weight of mixed refrigerant
- Requires multiple chillers and extremely large heat exchanger
- High energy consumption needed for effective liquefaction

4. Nitrogen Expansion

- Refrigeration by repetitive compression and expansion of gaseous refrigerant
- Low weight/space requirements and good design flexibility
- Requires implementation of multiple compressors
- Process is used for small scale LNG facilities

Problem

All technological cycles described above exhibit similar problems

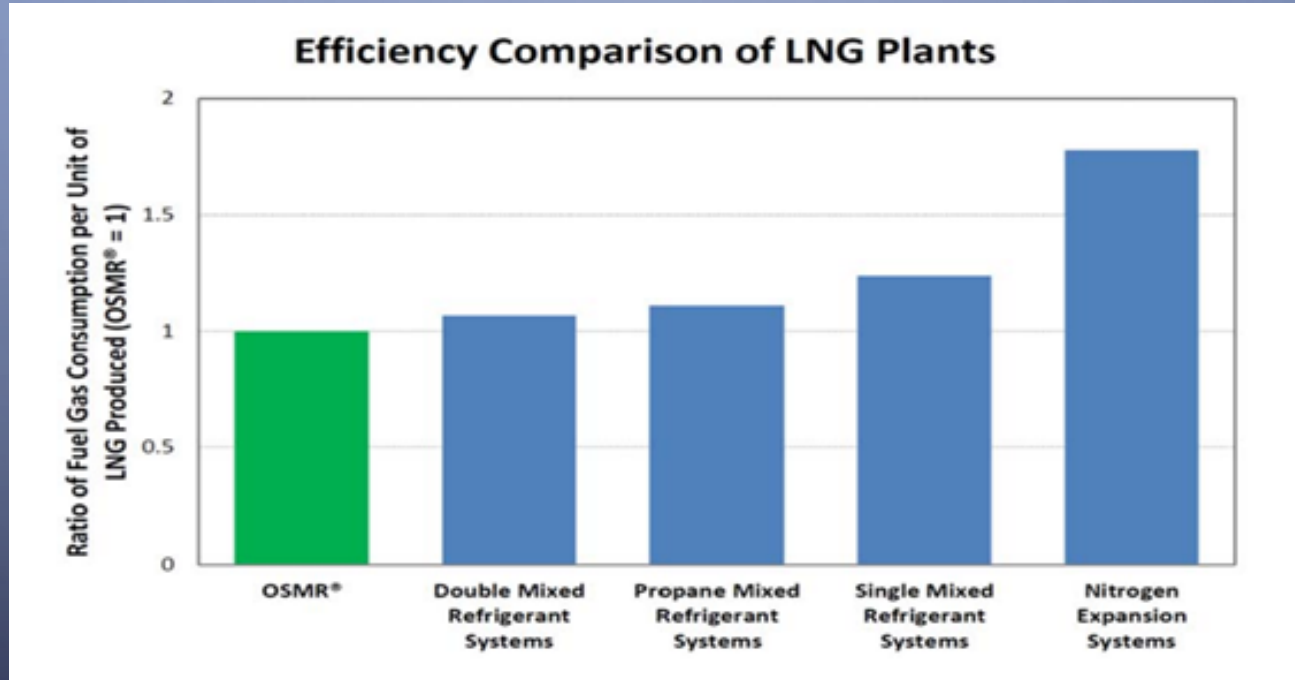
- High energy consumption
- Excessive methane emission from refrigeration and leak
- High CO₂ emission from power generation
- Low overall efficiency
- High equipments and operation costs

Possible Solution/ Competing Process

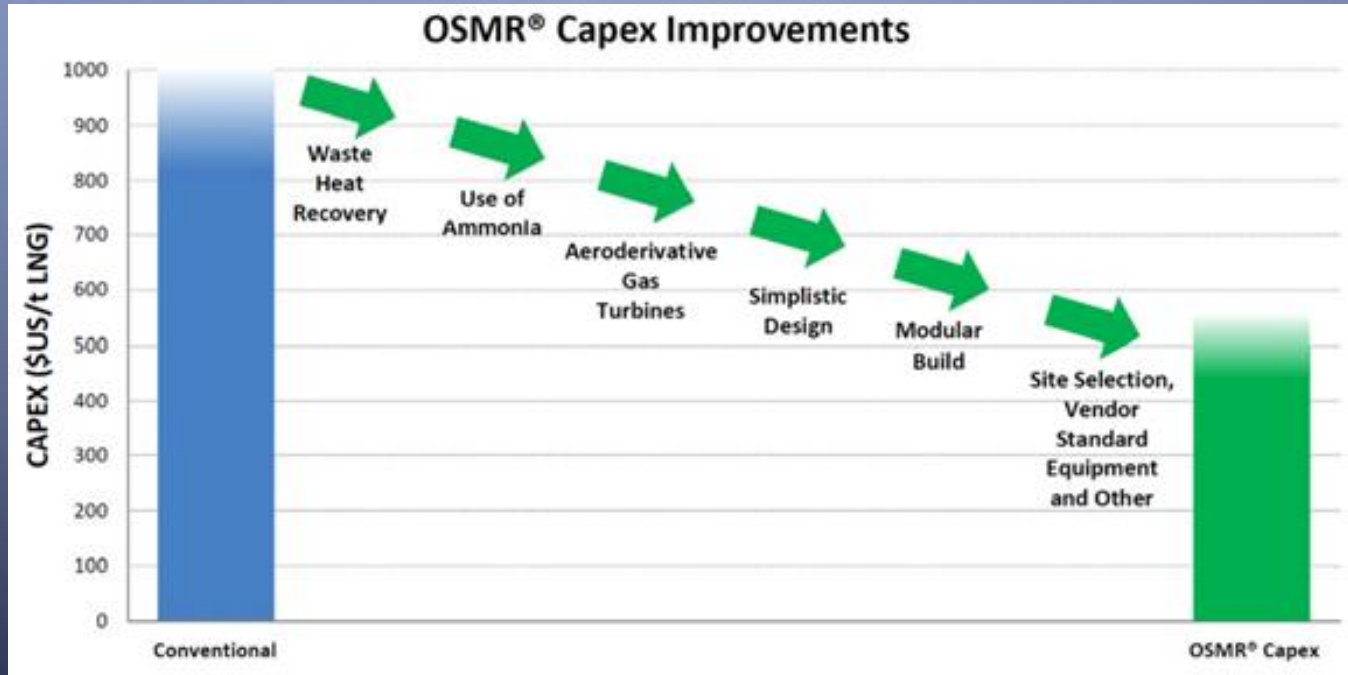
Optimized Single Mixed Refrigeration

- Uses Ammonia as a pre cooling refrigerant (increases plant capacity without causing price increase of major units)
- Combination of several well-proven, existing technologies into one integrated system
- Innovative, simple, reliable, low cost, highly efficient, environmentally friendly, robust and low risk
- Relatively simple to design, construct, operate and maintain

Efficiency Comparison of OSMR and Most Popular Cycles



Component capital cost of OSMR vs Conventional plant design (~US\$1,000/tonne)



Chosen Solution

OSMR exhibits the most efficient, cost-effective and environmentally friendly outcome.

Reasons:

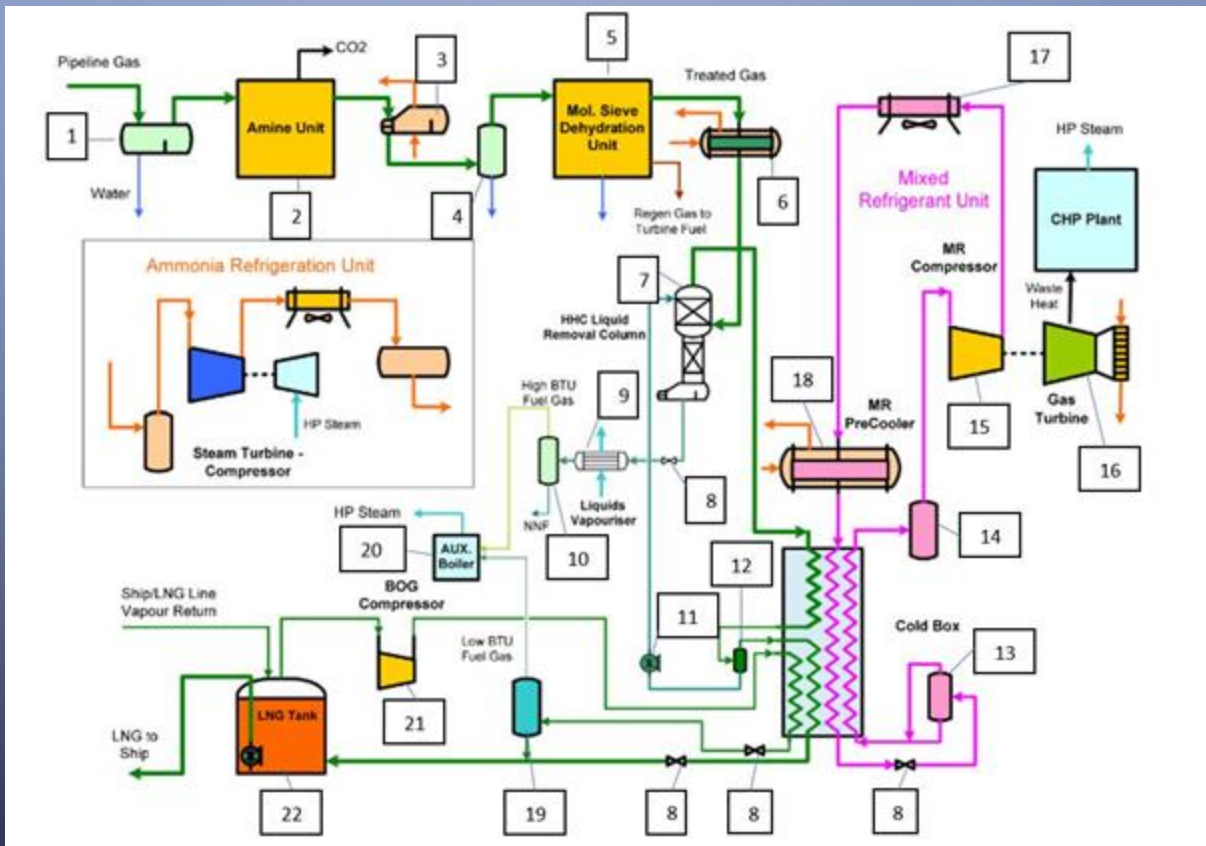
- Most waste heat is recovered and used as a means of regeneration power for compressors
- Offers extensive operation flexibility in terms of shifting refrigeration load
- Easy plant capacity scaling to meet customer requirements

DESIGN BASIS

- The Chevron oil refineries in California were producing and average of 25 million cubic feet of natural gas every day in 2018.
- To keep up with such large, growing demand for natural gas, our goal is to process and liquify 20 million cubic feet of natural gas every day, or 6.6 billion cubic feet of natural gas per year.
- All gases, liquids and equipment will be considered ideal.
- Separators will removed enough of the impurity to reach final product specifications

1. Chevron. "United States Highlights of Operations." Chevron.com, Chevron, 14 Aug. 2019, www.chevron.com/worldwide/united-states.

PFD:



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THANK YOU FOR
LISTENING!

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